WORKED SOLUTION

STEP I Specimen paper, Question 13

Mechanics • Circular Motion

STEP I Specimen paper, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q13

The key idea

This is a conical pendulum whose string can stretch. Hooke's law ties the tension to the length of the string, while resolving forces vertically and horizontally ties the tension to g and to ω. Because the string makes a non-zero angle with the vertical, the tension has to be strictly bigger than mg, and that single fact drives both inequalities.

Solution

Model the particle as a point mass on a light elastic string obeying Hooke's law, with no air resistance. Let λ be the modulus of elasticity.

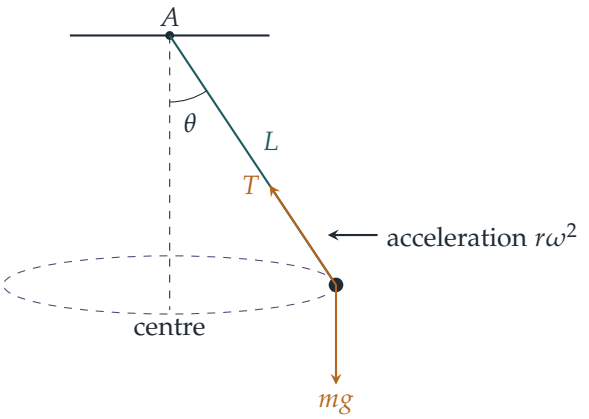
The modulus. Hanging at rest, the string has extension $2a - a = a$ and the tension balances the weight:

$$mg = \frac{\lambda a}{a} \quad \Longrightarrow \quad \lambda = mg.$$

So when the string has length L, its tension is

$$T = \frac{mg(L-a)}{a}.$$

The circular motion. Suppose the particle moves in a horizontal circle below A, with the string of length L making angle θ with the downward vertical. Since the motion is a genuine circle below the level of A, we have $0 < \theta < \frac{\pi}{2}$, so $\sin\theta > 0$ and $0 < \cos\theta < 1$. The centre of the circle is directly below A, and the radius is $r = L\sin\theta$.



The angular speed of the particle about the vertical through A is ω, so its acceleration is $r\omega^2 = L\omega^2 \sin\theta$, directed horizontally towards the centre. There is no vertical acceleration.

Resolving vertically:

$$T\cos\theta = mg. \tag{1}$$

Resolving horizontally, towards the centre:

$$T\sin\theta = mL\omega^2\sin\theta \quad \Longrightarrow \quad T = mL\omega^2, \tag{2}$$

where we may divide by $\sin\theta$ because $\sin\theta \neq 0$.

Ask yourself Where does the fact that the circle is genuine (not just the particle hanging still) get used?

It is used twice: to divide by $\sin\theta$ in (2), and to say $\cos\theta < 1$ below. If $\theta = 0$ the particle simply hangs at rest on the axis, (2) says nothing, and any ω would "work".

The string is longer than $2a$. From (1), since $0 < \cos\theta < 1$,

$$T = \frac{mg}{\cos\theta} > mg.$$

Using Hooke's law, this says

$$\frac{mg(L-a)}{a} > mg \quad \Longrightarrow \quad L - a > a \quad \Longrightarrow \quad L > 2a,$$

so the string is stretched to a length greater than $2a$.

The range of ω. Equating the two expressions for T, from Hooke's law and from (2):

$$mL\omega^2 = \frac{mg(L-a)}{a} \quad \Longrightarrow \quad \omega^2 = \frac{g}{a}\left(1 - \frac{a}{L}\right).$$

Since $L > 2a$, we have $0 < \frac{a}{L} < \frac{1}{2}$, so $\frac{1}{2} < 1 - \frac{a}{L} < 1$. Therefore

$$\frac{g}{2a} < \omega^2 < \frac{g}{a},$$

which is what we had to show.

The converse (not required, but reassuring). If ω lies in this range, set $L = \dfrac{a}{1 - a\omega^2/g}$. This is positive, satisfies $L > 2a$, and gives $\cos\theta = \dfrac{mg}{T} = \dfrac{a}{L-a}$, which lies strictly between 0 and 1. So for every such ω there really is a conical motion.

Checks

- Limits: as $L \to 2a$ the tension tends to mg, $\theta \to 0$ and $\omega^2 \to g/(2a)$, which is the frequency of small swings of a simple pendulum of length $2a$. As $L \to \infty$, $\omega^2 \to g/a$.
- Numerically: with $\omega^2 = \frac{3}{4}\,g/a$ we get $L = 4a$, $T = 3mg$ and $\cos\theta = \frac{1}{3}$. Stepping the full three-dimensional equations of motion from these starting conditions for two revolutions, the particle stayed on the predicted circle to within 10^{-13} of a unit.
- Units: g/a has units of s^{-2}, as ω^2 should.

Common slips

- Taking the radius of the circle to be L rather than $L\sin\theta$. The $\sin\theta$ then fails to cancel in (2).
- Forgetting that $T > mg$ needs $\cos\theta < 1$, that is, $\theta \neq 0$. The strict inequalities depend on it.
- Using the natural length a (or the equilibrium length $2a$) as the radius instead of the stretched length L.

Beyond STEP

The lower limit $\omega^2 = g/(2a)$ is the frequency of small oscillations about the vertical, and this is no accident: steady conical motion near the axis is just two perpendicular small swings, a quarter-period out of phase. The upper limit is where the string's stiffness can no longer supply the centripetal force, however far it stretches. Studying steady motions and their stability, for example by asking what happens if the particle is nudged off its circle, is a standard topic in first-year dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats the conical pendulum and the analysis of motion in rotating systems with care.

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